📘 BPSC CS Module 01: Digital Electronics & Number Systems (Sequence & Trick Master Notes)

> 🎯 **Target:** 15 / 15 Marks in Part-III (BPSC TRE Standard)
> **Rule:** No extra fluff. Pure sequence, exact steps, and memory formulas.

🔢 Sequence 1: Number Systems & Fast Conversion

1.1 Base & Bit Capacity

- **Binary (Base 2):** Digits `0, 1` \to **1 bit** per digit
- **Octal (Base 8):** Digits `0 to 7` \to **3 bits** per digit ($2^3 = 8$)
- **Decimal (Base 10):** Digits `0 to 9`
- **Hexadecimal (Base 16):** Digits `0 to 9, A(10), B(11), C(12), D(13), E(14), F(15)` \to **4 bits** per digit ($2^4 = 16$)

1.2 Binary \leftrightarrow Octal & Hexadecimal Grouping Rule

```
                    [ DECIMAL POINT . ]
      INTEGER PART                          FRACTIONAL PART
⟸ (Right to Left)                      (Left to Right) ⟹
Group from Point towards LEFT          Group from Point towards RIGHT
(Add 0 on extreme LEFT if needed)      (Add 0 on extreme RIGHT if needed)
```

📌 Step-by-Step Sequence:

1. **Octal ke liye: 3-3 bits** ke groups banayein.
2. **Hexadecimal ke liye: 4-4 bits** ke groups banayein.
3. **Point se pehle:** Point ke bagal se shuru karke **Left ki taraf (\leftarrow)** chalo. Kam pade to Left me `0` lagao.
4. **Point ke baad:** Point ke bagal se shuru karke **Right ki taraf (\rightarrow)** chalo. Kam pade to Right me `0` lagao.

- **Example 1 (Binary to Octal):** $(110101.11)_2$
- Point ke pehle (\leftarrow): `101` = **5**, `110` = **6** \implies **65**
- Point ke baad (\rightarrow): `11` me ek zero joda \to `110` = **6** \implies **.6**
- 👉 **Answer:** $(65.6)_8$
- **Example 2 (Binary to Hex):** $(10110111)_2$
- Right group (4 bits): `0111` = **7**
- Left group (4 bits): `1011` = 11 = **B**
- 👉 **Answer:** $(B7)_{16}$

1.3 Decimal Fraction \to Binary Sequence

- Number ko lagatar $\times 2$ karo.
- Har baar jo Integer (0 ya 1) aayega, use **Top to Bottom** likho:
- $0.625 \times 2 = \mathbf{1}.25$ (Carry 1)
- $0.25 \times 2 = \mathbf{0}.50$ (Carry 0)
- $0.50 \times 2 = \mathbf{1}.00$ (Carry 1 \to finish!)
- 👉 **Answer:** $(0.101)_2$

🔢 Sequence 2: 2's Complement 3-Second Inspection Rule

- **1's Complement:** Saari bits ulto (`0 ⇔ 1`).
- **2's Complement:** $\text{1's Complement} + 1$.

```
[ 1   0   1   1   0   0 ]
  |   |   |   |   └───┘ ── Rightmost 0s: Same rakho (00)
  |   |   |   └─────────── Pehla 1: Same rakho (1)
  |   └───┴─────────────── Iske baad Left ki saari bits INVERT kar do (010)
  Result: 0 1 0 1 0 0
```

- **Step 1:** Rightmost (LSB) se dekhna shuru karo.
- **Step 2:** Jab tak **pehla 1** na mile, tab tak saari bits **SAME** rakho.
- **Step 3:** Us pehle 1 ke baad Left side ki baaki saari bits ko **INVERT (0 ko 1, 1 ko 0)** kar do!

🔢 Sequence 3: n-Bit Ranges (Direct Formula Sequence)

Representation	Minimum to Maximum Range	Zero (0) ke roop	Total Distinct Values
Unsigned	0 **se** $+(2^n - 1)$	1 (`00 ... 0`)	2^n
Signed Magnitude	$-(2^{n-1} - 1)$ **se** $+(2^{n-1} - 1)$	2 (`+0` aur `−0`)	$2^n - 1$
1's Complement	$-(2^{n-1} - 1)$ **se** $+(2^{n-1} - 1)$	2 (`+0` aur `−0`)	$2^n - 1$
2's Complement	-2^{n-1} **se** $+(2^{n-1} - 1)$	**Sirf 1 (`00 ... 0`)**	2^n

- 🎯 **8-Bit Direct Numbers ($n = 8$):**
- 2's Complement range: -128 **se** $+127$
- Extra negative number: -128 (`10000000`)
- MSB weight in 2's complement: -2^{n-1} (Negative weight)

🔢 Sequence 4: Codes (BCD, Excess-3, Gray, Hamming)

1. **BCD (8421):**

- Valid digits: 0 **se** 9 (`0000` se `1001`).
- Invalid states: 10 **se** 15 (Total 6 invalid codes).
- **Addition Rule:** Agar sum > 9 ho ya carry out aaye $\implies +6$ **(`0110`)** add karo.

2. **Excess-3 (XS-3):**

- Formula: $\text{BCD} + 0011_2$ $(+3)$.
- **Self-Complementing:** Excess-3 ka 1's complement = Us number ka **9's complement**.

3. **Gray Code (Unit Distance Code):**

- Do consecutive numbers me **sirf 1 bit** ka difference hota hai.
- **Binary** $\to$ **Gray:** MSB same rakho ($G_3 = B_3$), fir consecutive XOR karo ($G_2 = B_3 \oplus B_2$).
- **Gray** $\to$ **Binary:** MSB same rakho ($B_3 = G_3$), fir diagonal XOR karo ($B_2 = B_3 \oplus G_2$).

4. **Hamming Code (Error Detection & Correction):**

- Detect s bit errors: $\mathbf{d_{\min} \ge s + 1}$
- Correct t bit errors: $\mathbf{d_{\min} \ge 2t + 1}$
- Parity bits (p) for m data bits: $\mathbf{2^p \ge m + p + 1}$

🔢 Sequence 5: Minimum Universal Gates Sequence (1, 2, 3, 4, 5)

Dono universal gates ke liye bas ye sequence yaad rakhiye:

```
NAND Gates Sequence:  NOT (1) ──> AND (2) ──> OR (3) ──> XOR (4) ──> XNOR (5)
NOR Gates Sequence:   NOT (1) ──> OR (2)  ──> AND (3) ──> XNOR (4) ──> XOR (5)
```

Target Gate	Min NAND Required	Min NOR Required
NOT	**1**	**1**
AND	**2**	**3**
OR	**3**	**2**
NOR	**4**	**1**
NAND	**1**	**4**
XOR	**4**	**5**
XNOR	**5**	**4**

🔢 Sequence 6: Adders & Multiplexers

1. **Half Adder:**
   - $\text{Sum} = A \oplus B$, $\text{Carry} = AB$
   - Hardware: **1 XOR + 1 AND** (ya **5 NAND**)
2. **Full Adder:**
   - $\text{Sum} = A \oplus B \oplus C_{\text{in}}$, $\text{Carry} = AB + BC_{\text{in}} + AC_{\text{in}}$
   - Hardware: **2 Half Adders + 1 OR Gate** (ya **9 NAND**)
3. **Multiplexer (MUX):**
   - Formula: $\mathbf{\text{Inputs} = 2^{\text{Select Lines}}}$
   - **Tree MUX shortcut:** 2×1 MUX se $N \times 1$ MUX banane ke liye hamesha $N - 1$ MUX lagenge (16×1 banane ke liye $16 - 1 = \mathbf{15}$).

🔢 Sequence 7: Flip-Flop Equations & Key Features

Flip-Flop	Inputs	Characteristic Equation	Key Feature to Remember
SR	S, R	$Q_{n+1} = S + R'\bar{Q}_n$	$S=1, R=1 \implies$ **Invalid / Forbidden**
JK	J, K	$Q_{n+1} = J\bar{Q}_n + K'Q_n$	$J=1, K=1 \implies$ **Toggle** ($Q_{n+1} = \bar{Q}_n$)
D	D	$Q_{n+1} = D$	**Data delay / Buffer** (Output = Input)
T	T	$Q_{n+1} = T \oplus Q_n$	$T=1 \implies$ **Toggle**; **Frequency Divider** ($\div 2$)

- **Race-Around Condition (in JK):** Jab $J=1, K=1$ ho aur clock pulse $t_p > t_{pd}$ ho.
- **Solution: Master-Slave JK Flip-Flop** ya Edge Triggering.

🎯 Top 5 BPSC Model Questions (Sequence Applied)

1. **Binary `(110101.11)₂` in Octal:** Point ke left 3-3 group (`110 101` $\to$ 65), right group (`.110` $\to$ .6) $\implies$ `65.6₈`.
2. **2's complement of `10110010`:** Right se pehla 1 mila bit 2 par (`...10`). Baaki left bits ulto $\implies$ `01001110`.

3. **8-bit 2's complement `11110000` in decimal:** MSB weight is $-128 \implies -128 + 64 + 32 + 16 = \mathbf{-16}$.

4. **Minimum NAND for XOR gate:** Sequence 1, 2, 3, 4 $\implies$ **4 NAND**.

5. **How many** 2×1 **MUX for** 16×1 **MUX:** $N - 1 = 16 - 1 = \mathbf{15}$.